

IgA nephropathy

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Abstract

IgA nephropathy (IgAN), the most prevalent primary glomerulonephritis worldwide, carries a considerable lifetime risk of kidney failure. Clinical manifestations of IgAN vary from asymptomatic with microscopic or intermittent macroscopic haematuria and stable kidney function to rapidly progressive glomerulonephritis. IgAN has been proposed to develop through a ‘four-hit’ process, commencing with overproduction and increased systemic presence of poorly O-glycosylated galactose-deficient IgA1 (Gd-IgA1), followed by recognition of Gd-IgA1 by antiglycan autoantibodies, aggregation of Gd-IgA1 and formation of polymeric IgA1 immune complexes and, lastly, deposition of these immune complexes in the glomerular mesangium, leading to kidney inflammation and scarring. IgAN can only be diagnosed by kidney biopsy. Extensive, optimized supportive care is the mainstay of therapy for patients with IgAN. For those at high risk of disease progression, the 2021 KDIGO Clinical Practice Guideline suggests considering a 6-month course of systemic corticosteroid therapy; however, the efficacy of systemic steroid treatment is under debate and serious adverse effects are common. Advances in understanding the pathophysiology of IgAN have led to clinical trials of novel targeted therapies with acceptable safety profiles, including SGLT2 inhibitors, endothelin receptor blockers, targeted-release budesonide, B cell proliferation and differentiation inhibitors, as well as blockade of complement components.

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Introduction

IgA nephropathy (IgAN) is one of the most common glomerulonephritides in the world and is a leading cause of kidney failure in individuals under the age of 40 years¹. The disease course of IgAN is highly heterogeneous. The spectrum includes mild forms in which spontaneous remission is conceivable, slowly progressive forms and rapidly progressive forms in which kidney function decreases rapidly, leading to kidney failure and the need for dialysis². Overall, life expectancy is modestly reduced for patients with IgAN but is highly dependent on whether the disease progresses and leads to kidney failure³.

Most commonly asymptomatic, IgAN is usually diagnosed following an investigation for persistent proteinuria and haematuria. Over a period of 20–30 years, as many as 30–40% of adult patients with IgAN progress to kidney failure^{2,4,5}.

The pathogenetic mechanisms of IgAN remain incompletely understood. IgA is the predominant class of immunoglobulins, representing nearly 70% of the body's total immunoglobulin. IgA is primarily present in mucous secretions and has a pivotal role in providing local immunity against infections, tolerance to harmless environmental pathogens and development of the microbiome. It is believed that the development of IgAN originates from aberrant glycosylation of O-linked glycans of IgA1, resulting in galactose-deficient IgA1 (Gd-IgA1)⁶. Circulating immune complexes containing IgG, IgA or IgM autoantibodies bound to polymeric Gd-IgA1 then deposit within the glomerular mesangium, leading to the proliferation of mesangial cells, glomerular matrix deposition and immune cell infiltration in the kidney⁷. The exact reasons why the kidney is primarily affected by this systemic disease are still largely unclear. Insights into the influence of the mucosal immune system in the pathogenesis of IgAN include accumulating evidence for a dysregulation of mucosal IgA production. Factors important for the development of plasma cells, including IgA-producing plasma cells, such as B cell activation factor (BAFF) and a proliferation-inducing ligand (APRIL), seem to be involved, as well as aberrant trafficking of IgA-producing plasma cells in the human body^{8,9}.

In this context, new milestones for the therapy of IgAN have been set in the past 5 years. The findings of the NefIgArd trial published in 2023 (ref. 10) provide strong evidence for the involvement of the intestinal mucosal immune system as the site of highest IgA production in the body. In this trial use of the targeted release formulation budesonide (TRF-budesonide) that is released preferentially in the terminal ileum (a site of high IgA production) reduced the activity parameters of IgAN. Other emerging and future treatment options for IgAN include sodium–glucose co-transporter 2 (SGLT2) inhibitors and dual endothelin A blockers^{11,12}. In addition, ongoing clinical trials are investigating further treatment options, including plasma cell maturation and complement inhibition.

In this Primer, we summarize the epidemiology of IgAN, as well as current advances in understanding the pathophysiology of the disease. We discuss the clinical presentation, diagnosis and biomarkers of IgAN and treatment options. Finally, we review data on the quality of life of patients with this disease, and identify gaps in current knowledge and make recommendations for future research efforts.

Epidemiology

The prevalence of IgAN varies geographically with the highest prevalence in the Asian Pacific region and the lowest in Africa (Fig. 1). The reported prevalence varies based on the primary source of data. Biopsy registry data usually underestimate the prevalence of IgAN as patients with mild disease often do not undergo biopsy. IgAN has a reported

prevalence of >2.5 cases per 100,000 in Europe and North America¹³. Registry data indicate that IgAN is the most common biopsy-proven primary glomerular disease in China, present in up to 54% of all glomerulonephritis cases, and these data also suggest that IgAN is the most common cause of chronic kidney disease (CKD) in China¹⁴.

Geographical variation in IgAN prevalence can be partially explained by variations in the accessibility of primary care, with the poorest access in Africa and the greatest access in Asia, Europe and North America¹⁵. In addition, the decision to perform or not perform a kidney biopsy considerably affects the reported prevalence of IgAN. Overall, a lower clinical threshold to perform a kidney biopsy results in a greater number of IgAN diagnoses. In Africa, very low frequencies of IgAN have been reported with a minimum of 0.8% of biopsy-proven primary glomerular disease, which probably reflects both accessibility to kidney biopsy and a much higher clinical threshold to undertake an expensive investigation¹⁶. However, there is evidence that even beyond differences in medical care, the prevalence of IgAN may be lower in individuals of African origin. In a small biopsy database analysis from the USA, fewer cases of IgAN were found in African Americans than in white Americans¹⁷.

In addition to biopsy data, geographical variation in IgAN prevalence is supported by data on the frequency of IgAN risk alleles, with the highest number of risk alleles present in individuals from East Asia and the lowest number in African and African American populations¹⁸. Associations with the major histocompatibility complex have been reported in genome-wide association (GWA) studies in Europeans and East Asians^{19–21}. A large bi-ethnic GWA study included individuals with IgAN of European and East Asian ancestry and revealed a strong west-to-east gradient of increasing frequency of risk alleles, consistent with the reported prevalence of IgAN. Intriguingly, the distribution of these risk alleles strongly correlates with local pathogen diversity suggesting that perhaps the mucosal immune system in people from East Asia has evolved to more effectively deal with prevalent pathogens in early life, which, as a consequence, has resulted in an increased risk of long-term development of IgAN and kidney failure in later life²².

In the most recent GWA study published in 2023, 16 new loci were identified²³ and, overall, GWA studies have identified almost 30 independent risk alleles for IgAN, explaining around 11% of the disease risk. Convergence of biological candidates on a common set of inflammatory signalling pathways and cytokine ligand–receptor pairs, prioritizing potential new drug targets, has been observed²³. However, owing to the high prevalence of subclinical IgAN in East Asians and the substantially increased contribution of East Asians to nearly all published GWA studies, a concern remains that a considerable proportion of the control populations for these studies may also have IgAN.

The male-to-female ratio of IgAN has been reported as 3:1 in Europe and 1:1 in East Asia²⁴. The reason for this discrepancy is currently not understood. Diagnosis of IgAN is most common in young adults and children, and often follows a presentation with visible haematuria following an upper respiratory tract infection. In adults, the median age at diagnosis is 40–45 years in most Western series; however, adult diagnoses are often made earlier in countries with an active urine screening programme, such as Japan^{24–26}.

Mechanisms/pathophysiology

The widely accepted multi-hit hypothesis of the pathogenesis of IgAN states that multiple factors, such as genetic predisposition, abnormal galactosylation of IgA1 and formation of IgA-containing immune complexes, as well as mucosal infections or other environmental triggers,

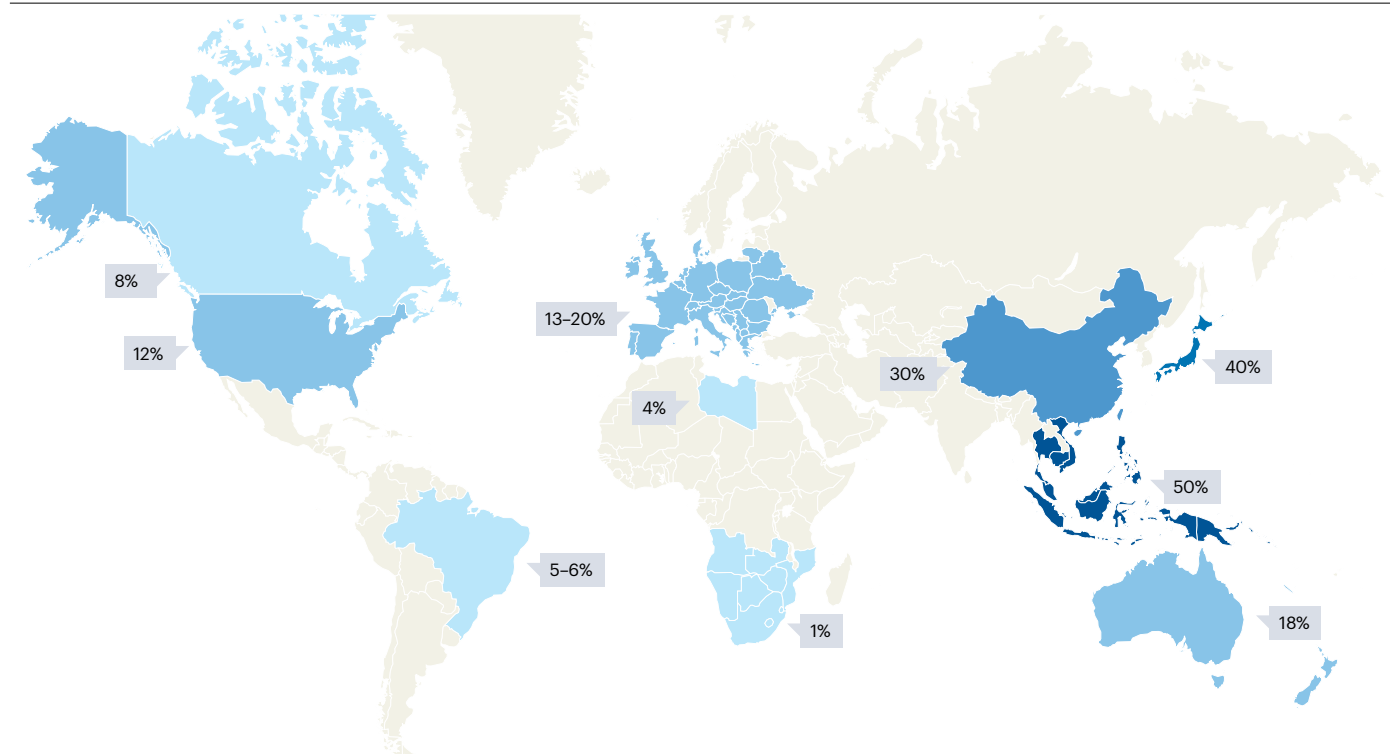


Fig. 1 | Prevalence of IgAN worldwide. Percentages of patients with glomerular disease who have IgA nephropathy (IgAN). IgAN is the predominant glomerular disease in resource-abundant countries worldwide, and displays

notable geographical variation. The disease is most prevalent in East Asian and white populations, but is relatively uncommon among Black individuals. Data from ref. 285.

jointly trigger the development and progression of IgAN⁷ (Fig. 2). Animal models were essential in understanding the pathogenesis of IgAN (Box 1).

Pathogenic IgA

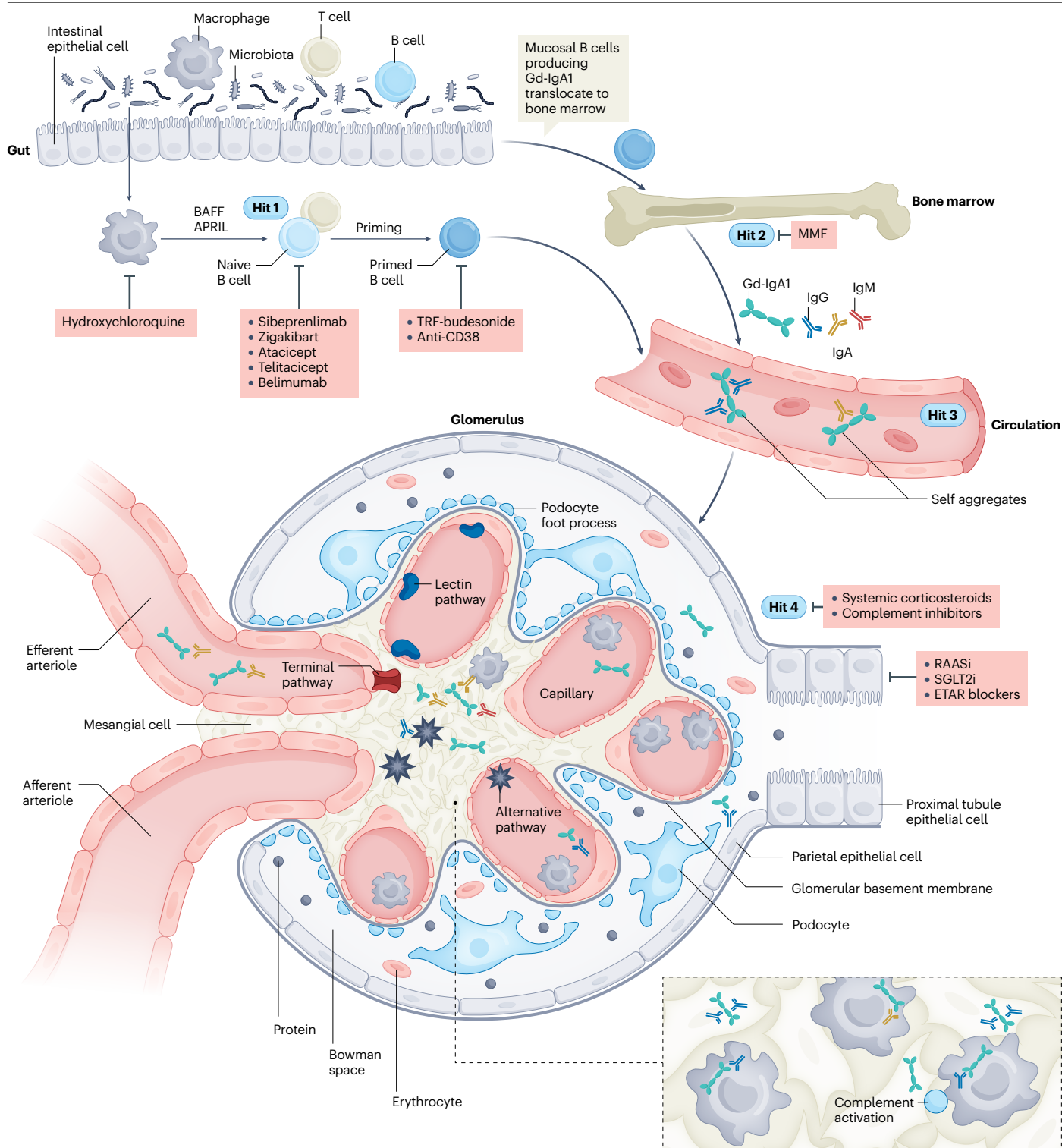
In IgAN, increased amounts of aberrantly glycosylated IgA1 (Gd-IgA1) are present in patient serum and glomerular immune deposits²⁷⁻³⁰. These Gd-IgA1 molecules, are produced by IgA1-secreting cells, typically normal functioning B cells. The pathogenetic changes comprise reduced galactosylation of O-linked *N*-acetylgalactosamine residues with or without altered sialylation in the IgA1 hinge region³¹. These alterations, and specifically IgA1 O-glycan sialylation, are closely linked to disease activity and kidney function in IgAN³².

IgA exists in two isoforms, IgA1 and IgA2, and both isoforms can exist as monomers and polymers. Unlike IgA2, IgA1 has hinge region comprising 18 amino acids in which 3-6 O-glycans are attached to some of the serine and threonine residues³³. IgA1 O glycosylation involves an initial addition of *N*-acetylgalactosamine (GalNAc) by a GalNAc-transferase (GalNAc-T); the monosaccharide structure of GalNAc linked to a serine or threonine is known as the Tn antigen. This step may be followed by the addition of galactose to GalNAc by the enzyme core 11,3-galactosyltransferase (C1GalT1) and its chaperone, C1GalT1C1 (COSMC)³³. The galactose and/or GalNAc residues may be sialylated with a α -2,3-linked and α -2,6-linked sialic acid, respectively. In IgAN, some GalNAc residues are sialylated early, preventing subsequent additions of galactose³⁴. Altered glycan moieties, particularly with reduced galactose and sialic acid content, are observed in IgA1

molecules from patients with IgAN, and are associated with disease activity and reduced glomerular function³².

Alterations in the expression and/or activity of several glycosylation enzymes have been associated with an increased production of Gd-IgA1 (refs. 6,35-38). However, a universal defect in glycosylation enzymes in all B cells is unlikely to be the cause of decreased O glycosylation in IgAN, as IgD (which is also heavily O galactosylated) shows normal glycosylation in IgAN. Epigenetic control of IgA1 O glycosylation is thought to be important. Two microRNAs (miRNAs), miR-148b and miR-let-7b, which target mRNA for *C1GALT1* and *GALT2*, respectively, are upregulated in patients with IgAN compared with expression levels in healthy individuals, and their abnormal expression is associated with altered expression of the respective glycosyltransferases^{39,40}. In addition, expression of these miRNAs is considerably upregulated in white populations with IgAN compared with Asian populations, which may explain the higher Gd-IgA1 levels in patients of European descent⁴¹.

Other studies have identified new mechanisms involving competing non-coding RNAs in the regulation of IgA1-glycosylation in IgAN⁴². Two qualitative trait GWA studies have associated serum levels of Gd-IgA1 with single-nucleotide polymorphisms in the non-coding region of *C1GALT1* (refs. 22,43). These data suggest that changes in IgA1 O glycosylation may result from changes in the transcriptional regulation of *C1GALT1*, possibly in specific microenvironments such as the mucosa. These studies also offer a genetic explanation for the previously reported heritability of IgA1 O glycosylation⁴⁴. Of note, this association was established after neuraminidase treatment of serum IgA and an effect on sialylation of the IgA1 hinge region can, therefore, not be



excluded. Despite the higher prevalence of IgAN in Chinese populations, *C1galt1* risk haplotypes are less common among Chinese patients with IgAN than in patients of European descent, reflected by lower serum levels of Gd-IgA1 in Chinese patients⁴³, suggesting that there is not a simple relationship between serum Gd-IgA1 and IgAN.

Mucosa-kidney axis

Mucosa-associated lymphoid tissue (MALT) describes lymphoid tissue scattered along mucosal sites throughout the body, consisting of lymphocytes, plasma cells, dendritic cells and macrophages, and has a central role in the immune response to various antigens. The MALT

Fig. 2 | The multi-hit hypothesis of IgAN and therapeutic targets. The gut microbiome stimulates mucosal innate immune cells via Toll-like receptors, leading to excessive activation of mucosal IgA-committed B cells through B cell activation factor (BAFF) and a proliferation-inducing ligand (APRIL) signalling. The first hit involves the excessive generation of galactose-deficient IgA1 (Gd-IgA1), which enters the circulation from mucosal sites, or from systemic sources owing to mis-homing of mucosal B cells. The second hit involves the formation of anti-Gd-IgA1 antibodies (IgG, IgA or IgM), including self-aggregates. The formation of circulating immune complexes containing IgG, IgM or IgA autoantibodies bound to Gd-IgA1 represents the third hit. The fourth hit occurs when these circulating immune complexes deposit in the glomerular mesangium. This triggers mesangial cell activation, complement activation,

and recruitment of monocytes, macrophages and T cells. This process leads to podocyte injury and proteinuria, which fuels the ongoing inflammation. Local complement activation has a pivotal role in IgA nephropathy (IgAN) pathophysiology, with the alternative and lectin pathway being particularly relevant. Irrespective of the activation route, the common effector mechanism involves the generation of anaphylatoxins C3a and C5a, as well as the formation of the membrane attack complex C5b–9, leading to subsequent inflammation and injury. Displayed drug targets are based on laboratory evidence specifically in IgAN, but the targeting drugs might have effects at other sites as well. ETAR, endothelin A receptor; MMF, mycophenolate mofetil; RAASi, renin–angiotensin–aldosterone system inhibition; SGLT2, sodium–glucose co-transporter 2; TRF–budesonide; targeted release formulation budesonide.

distributes over the surfaces of all mucosal tissues and is divided into nasopharynx-associated lymphoid tissue (NALT) and gut-associated lymphoid tissue (GALT).

Although the exact origin of Gd-IgA1 deposits in the kidney remains elusive, a connection between IgAN and the MALT is likely, given the close association between visible haematuria and upper respiratory tract infections in patients with IgAN, pointing to a role of NALT in IgAN development⁴⁵. In individuals with IgAN, mesangial deposits of IgA1 consist predominantly of polymeric Gd-IgA1, which is characteristic of mucosal IgA1. By contrast, systemic IgA in healthy individuals is typically monomeric, heavily galactosylated and derived from bone marrow-residing plasma cells^{34,46}. Furthermore, in IgAN, the number of IgA plasma cells in the GALT in the duodenum decreases, while their number increases in systemic sites, notably the bone marrow^{47,48}. Hence, one intriguing hypothesis is that the increase in Gd-IgA1 in IgAN may be due to the misguided release of IgA with mucosal characteristics into the circulation instead of at mucosal surfaces and that this misdirected secretion results from mucosa-derived plasma cells that migrate to the bone marrow instead of returning to mucosal surfaces⁴⁸. Indeed, there is some evidence for defective homing receptor expression on B cells in IgAN, but more research is required to precisely define the pattern of B cell mistrafficking^{34,48–51} (Fig. 2). NALT and GALT are the MALTs that have been most studied in IgAN.

NALT. Several studies have documented elevated proportions of IgA⁺ cells and lower expression of glycotransferases in the tonsils of patients with IgAN⁵². Tonsillectomy performed 1 year after kidney transplantation owing to IgAN decreased the serum level of Gd-IgA1 and recurrence of histological IgAN after transplantation⁵³. However, the benefit of tonsillectomy in IgAN in non-transplant patients is inconsistent and tonsillectomy is, therefore, not routinely recommended⁵⁴.

Various studies have assessed the potential role of specific pathogens in stimulating the NALT in IgAN. Glomerular deposition of streptococcal M proteins has been reported in IgAN and IgA vasculitis (IgAV)⁵⁵. Furthermore, microbiome studies have revealed similarities in tonsillar microbiomes between patients with IgAN and those with frequent tonsillitis⁵⁶, as well as differences between patients with IgAN, patients with other glomerular diseases and healthy individuals⁵⁷. However, other studies have been unable to confirm this³⁴.

GALT. Compared with the NALT, more extensive evidence supports a pathogenetic link between the GALT and IgAN. The incidence of IgAN was found to be associated with gastrointestinal disorders, including inflammatory bowel disease and coeliac disease, and GWA studies have identified multiple risk alleles for IgAN associated with the gut

immune response, epithelial barrier integrity and inflammatory bowel disease^{19,22,23,58–60} (Table 1). The faecal microbiome composition may also influence IgA production and a reduced microbial diversity has been found in patients with progressive IgAN⁶¹. However, it is unclear whether this reduced microbial diversity is specific for IgAN or rather represents a more general feature of progressive CKD⁶². Finally, serum levels of BAFF are increased in IgAN, and BAFF levels positively correlate with the levels of five specific microbiota metabolites⁶³.

Innate immune components. Activation of the innate immune response appears central in the pathogenesis of IgAN. Toll-like receptors (TLRs) link innate and adaptive immunity⁶⁴. They are expressed on dendritic cells, B cells and macrophages, as well as on fibroblasts and

Box 1

Animal models of IgAN

Animal models have been a useful tool for studying aspects of the pathogenesis of IgA nephropathy (IgAN), despite the pronounced differences in IgA molecules, receptors and immune responses between humans and rodents. Most widely used is the ddY mouse, which is a spontaneous model of IgAN²⁸⁶. To overcome their extreme variability in disease onset and severity, a new mouse strain called grouped ddY was obtained by intercrossing an early-onset group of ddY mice for over 20 generations^{286,287}. Grouped ddY mice consistently develop IgAN-like disease within 8 weeks. The mechanisms leading to IgAN development involve abnormal glycosylation of IgA, similar to human IgAN. The mice produce deglycosylated mouse IgA, which forms immune complexes and deposits in the glomeruli, leading to kidney inflammation and damage. In addition, transgenic mouse models including the uteroglobin Tg model, Bcl2 Tg model, LIGHT Tg model, BAFF Tg model⁷¹ and β 4GalT-I KO model have been used to investigate the roles of specific genes in immune regulation in IgAN^{286,287}. Humanized mouse models expressing human CD89 and human IgA1, named α K1CD89Tg, have also been used to evaluate specific pathogenetic pathways and new treatment approaches using drugs and dietary and microbiome modification^{286,287}. Finally, the administration of exogenous antigens or viral challenges can induce IgAN-like phenotypes, and these have been used to study the early process of mesangial IgA deposition²⁸⁸.

Table 1 | Risk allele loci in IgAN

Susceptible loci in IgAN	Associated protein	Protein function	Examples of associated diseases
FCRL3	Fc receptor-like protein 3	Member of the immunoglobulin receptor superfamily; contains immunoreceptor-tyrosine activation motif (ITAM) and immunoreceptor-tyrosine inhibitory motif (ITIM); activation properties of FCRL3 were observed in relation to TLR9-mediated signalling	Mutations have been discovered in rheumatoid arthritis, systemic lupus erythematosus and autoimmune thyroid disease ²⁴⁴
TNFSF4	TNF superfamily member 4 (OX40L, CD252)	Is present on the surface of activated T lymphocytes (CD4 ⁺), NK cells, NKT cells and neutrophils	Polymorphisms were reported for systemic lupus erythematosus ²⁴⁶
TNFSF18	TNF superfamily member 18	Modulates T lymphocyte survival via its receptor ²⁴⁵	
CFH (SNP rs6677604) and CFH (SNP rs12029571)	Complement factor H	Complement control protein of the alternative pathway of complement	IgAN, atypical haemolytic uraemic syndrome and age-related macular degeneration ²⁴⁷
REL	REL	Member of the NF- κ B family of transcription factors; myeloid checkpoint protein ²⁴⁸	B cell lymphomas (diffuse large B cell lymphoma and Hodgkin lymphoma)
CD28	CD28	Receptor for CD80 and CD86, constitutively expressed on naive T cells; primary costimulatory molecule for T cell activation ²⁴⁹	Role in autoimmune conditions ²⁴⁹
PF4V1	Platelet factor 4 variant 1 (CXCL4L1)	Chemokine that is highly similar to platelet factor 4; the encoded protein has a strong antiangiogenic function and is regulated by C-X-C chemokine receptor 3	Inhibitor of angiogenesis; inhibitor of endothelial cell chemotaxis
IRF4/DUSP22	Interferon regulatory factor 4 (IRF4 or MUM1)	Transcription factor; has a role in B cell (also long-lived plasma cell) and T cell differentiation ^{250–252}	Acute leukaemia and large B cell lymphoma ^{253,254}
LY86	Lymphocyte antigen 86	May cooperate with CD180 and TLR4 to mediate the innate immune response to bacterial lipopolysaccharide (LPS) and cytokine production; important for efficient CD180 cell surface expression	None known
HLA-DRA	Major histocompatibility complex, class II, DR- α	α -Chain of antigen-presenting major histocompatibility complex class II (MHCII) molecule	CLL, diffuse large B cell lymphoma, colorectal cancer and lung cancer
HLA-DQB/DQA (SNP rs9275355), HLA-DQA and HLA-DQB/DQA (SNP rs9275596)	Major histocompatibility complex, class II, DQ- α and DQ- β	Central role in the immune system by presenting peptides derived from extracellular proteins	Diabetes ²⁵⁵ , coeliac disease ²⁵⁶ and multiple sclerosis ²⁵⁷
HLA-DPA and HLA-DPB	Major histocompatibility complex, class II, DP- α and DP- β	Central role in the immune system by presenting peptides derived from extracellular proteins	Diabetes ²⁵⁸
DEFA1 and DEFA4	Defensin α 1 and defensin α 4	α -Defensin family of antimicrobial peptides; defensins are expressed in granules of neutrophils for host defence	None known
LYN	LYN proto-oncogene	Scr-like tyrosine protein kinase; has an important role in the regulation of B cell differentiation, proliferation, survival and apoptosis, and is important for immune self-tolerance; acts downstream of several immune receptors, including the B cell receptor, CD79A, CD79B, CD5, CD19, CD22, FCER1, FCGR2, FCGR1A, TLR2 and TLR4	None known
ANXA3	Annexin A3	Member of the annexin binding family; regulation of cellular growth; inhibition of phospholipase A2	Cancer ²⁵⁹
TNFSF8	TNF superfamily member 8 (CD153)	Ligand of the TNF receptor CD30 (refs. 260,261); antimicrobial immune defence; expression on T _H 2 cells	None known
TNFSF15	TNF superfamily member 15 (also known as vascular endothelial growth inhibitor, VEGI)	TNFSF15 is expressed in endothelial cells (not in immune cells)	Inflammatory bowel disease ²⁶²
CARD9	Caspase recruitment domain-containing protein 9	Association with the CARD domain of BCL10 (NF- κ B activation) ²⁶³ ; important in antifungal, innate immune defence ²⁶⁴	Ankylosing spondylitis and inflammatory bowel disease ²⁶⁵
REEP3	Receptor accessory protein 3	Structure and function of the endoplasmic reticulum	Neurological diseases, juvenile idiopathic arthritis, cardiac diseases and retinal diseases ²⁶⁶
ZMIZ1	Zink finger MIZ domain-containing protein 1	Acts as a transcriptional coactivator	Locus associated with psoriasis ²⁶⁷ ; prognostic marker in renal cancer (favourable) ²⁶⁸

Table 1 (continued) | Risk allele loci in IgAN

Susceptible loci in IgAN	Associated protein	Protein function	Examples of associated diseases
OVOL1	Ovo like transcriptional repressor 1	Can modulate cellular plasticity ²⁶⁹ and mesenchymal to epithelial transition	Cancer ²⁷⁰
RELA	Transcription factor p65	Involved in NF-κB signalling	Behçet-like autoinflammatory disease 3 (ref. 271)
ETS1	Protein C-ets-1	ETS family of transcription factors	Immune cell function and cancer ²⁷²
IGH	Immunoglobulin heavy chain	None known	None known
ITGAM and ITGAX	Integrin αM (CD11b) and Integrin αX (CD11c)	Transmembrane proteins on B cells, dendritic cells, NK cells monocytes, macrophages and neutrophils	Systemic lupus erythematosus ²⁷³
IRF8	Interferon regulatory factor 8	Transcription factor, regulates expression of genes stimulated by type I interferons; myeloid cell differentiation ²⁷⁴	Chronic myelogenous leukaemia ²⁷⁵
TNFSF12	TNF superfamily member 12 (also known as TNF-related weak inducer of apoptosis (TWEAK))	Cytokine on leukocytes ²⁷⁶	Inflammation, fibrosis and angiogenesis ²⁷⁷
TNFSF13	TNF superfamily member 13 (also known as A proliferation-inducing ligand (APRIL))	Cytokine important for B cell signalling ²⁷⁸	Autoimmunity and IgAN ²⁷⁹
TNFRSF13B	TNF receptor superfamily member 13B (TACI)	Receptor for BAFF and APRIL, mainly on B cells	Myeloma, CLL, diffuse large B cell lymphoma and acute myeloid leukaemia ²⁸⁰
FCAR	Transmembrane receptor FcαRI (CD89)	FcαRI binds the heavy-chain constant region of IgA antibodies ²⁸¹	None known
LIF	Leukaemia inhibitory factor	Pleiotropic cytokine of IL-6 superfamily; induces differentiation in myeloid leukaemia cells and inhibits their proliferation; has a role in stem cells (homeostasis and regeneration of tissues); regulatory role in immunity ²⁸² ; signals via gp130	Cancer, cachexia
OSM	Oncostatin M	Signals via gp130	Inflammation, autoimmunity, cancer and arteriosclerosis
C1GALT1	Core 1 synthase, glycoprotein-N-acetylgalactosamine 3-β-galactosyltransferase 1	Mucin type O glycosylation; angiogenesis, oncogene	Cancer ²⁸³

CLL, chronic lymphocytic leukaemia; IgAN, IgA nephropathy; NK, natural killer; NKT cell, natural killer T cell; SNP, single-nucleotide polymorphism; T_H2, T helper 2.

epithelial cells. TLR activation through pathogen-associated molecular patterns stimulates B cell differentiation, leading to increased IgA production and contributing to inflammatory processes. In patients with IgAN, expression of *TLR4* and *TLR7* mRNAs is increased in peripheral blood mononuclear cells compared with cells from healthy individuals, and TLR7 in B cells is associated with increased serum levels of Gd-IgA1 and inflammation markers^{65,66}. Elevated expression of *TLR4*, *TLR7*, *TLR8* and *TLR9* has also been detected in IgAN kidney biopsy samples and this expression is correlated with reduced estimated glomerular filtration rate (eGFR) and increased proteinuria⁶⁷. Similarly, high serum levels of TLR9 were associated with increased serum IgA1 levels, disease activity and kidney function decline over time⁶⁸. In addition to driving mucosal IgA production, TLR activation can also modify glycosyltransferase activity by altering the methylation of the *COSMC* gene, resulting in decreased *C1GALT1* expression⁶⁹.

Adaptive immunity components. BAFF and APRIL are critical for B cell maturation, IgA class switch recombination and plasma cell survival, and have been implicated in various autoimmune diseases^{48,70}.

Transgenic mice overexpressing BAFF display an IgAN-like kidney phenotype⁷¹. Tonsillar mononuclear cells from patients with IgAN exposed to the TLR9 agonist CpG-ODN produce high levels of BAFF and IgA, and blocking of BAFF signalling reduces this IgA response⁷². In vitro stimulation of IgAN tonsillar mononuclear cells, either through vibratory stimulation or with capsaicin, upregulates BAFF synthesis and suppresses *C1GALT1* and *COSMC* mRNA expression, increasing the secretion of Gd-IgA1 (refs. 73,74). Elevated serum BAFF levels are associated with worse kidney histopathology and higher serum creatinine, a serum marker of kidney damage^{68,75}, but it is uncertain whether this is a causal relationship or represents non-specific retention of BAFF in advancing kidney failure.

GWA studies in IgAN have identified *TNFSF13* (which encodes APRIL) as a susceptibility locus, with the risk variant also being associated with increased total IgA serum levels⁵⁸. The expression of APRIL and its receptors are elevated in B cells in IgAN and are associated with Gd-IgA1 secretion⁹. Tonsillar TLR9 and APRIL levels are also elevated in IgAN, and both correlate with proteinuria⁷⁶. APRIL has also been shown to increase IgA synthesis by tonsillar mononuclear cells⁷⁷ and an

APRIL-specific small interfering RNA abolished IL-6-mediated Gd-IgA1 synthesis in human IgA1-secreting cells⁷⁸.

Immune complex formation. Current evidence shows that poorly glycosylated IgA1 O glycoforms tend to self-aggregate and to form antigen–antibody complexes with IgG, IgA and IgM antibodies, leading to macromolecular immune complexes that are prone to deposit in the mesangium^{6,79,80} (Fig. 2). Autoantibodies specific for the poorly O glycosylated hinge region of IgA1 (refs. 79–81) have been variably reported to be associated with disease activity^{82,83}. Gd-IgA1–IgG immune complexes injected into immunodeficient mice deposit in the mesangium together with murine complement C3 and trigger glomerular injury similar to that seen in IgAN, whereas injection of the individual components (IgA1 or IgG) has no such effect^{84,85}. One study identified serum immune complexes containing predominantly IgM and to a lesser extent IgG and IgA, which are elevated in patients with IgAN and induce proliferation of human mesangial cells in vitro⁸⁶. Importantly, the presence of IgG and/or IgM are not required for a diagnosis of IgAN. In most kidney biopsy studies co-deposition of IgG and IgM is found in only a few patients, raising doubts about the importance of IgG and IgM in the pathogenesis of IgAN. Circulating IgA immune complexes may also contain various other proteins, many of which are likely to have considerable biological effects when deposited in the mesangium⁸⁷.

Soluble CD89 (sCD89, also known as Fc fragment of IgA receptor), which is generated following IgA-dependent shedding from the surface of myeloid cells, is also implicated in the pathogenesis of IgAN^{88–90}. Transgenic mice expressing human CD89 develop circulating sCD89–IgA complexes, resulting in mesangial IgA deposition, haematuria and proteinuria. Remarkably, infusing serum containing sCD89–Gd-IgA complexes from these transgenic mice into wild-type mice induces similar renal disease. By contrast, IgA from patients with IgAN triggers mesangial IgA deposition in the transgenic mice but has no effect in wild-type mice, similar to IgA from healthy individuals⁸⁹. Moreover, other studies found an increase in serum sCD89–IgA complexes in children and adults with IgAN^{91–93}. These complexes and free sCD89 have also been seen in mesangial deposits in biopsy samples from patients with IgAN in some studies. Of note, both sCD89–IgA complexes and sCD89 alone can directly induce mesangial cell proliferation in vitro and in vivo⁹². These complexes can bind mesangial transferrin receptor (CD71) through interactions with both sCD89 and IgA1 (ref. 94). Upon deposition on mesangial cells, sCD89 and sCD89–IgA1 complexes induce surface expression of transglutaminase 2 (TGASE2)⁹², which is overexpressed in the mesangium of patients with IgAN and correlates with renal function decline⁹⁵. In vitro, sCD89 binds to CD71, inducing mesangial proliferation via the mTOR pathway, which is inhibited by mTOR blockers providing potential new therapeutic avenues for proliferative IgAN. Finally, sCD89 and TGASE2 have also been found as biomarkers in the urine of patients with IgAN^{91,96}.

Mesangial accumulation of IgA immune complexes is a prerequisite for a diagnosis of IgAN, but the pathological consequences of this deposition are highly variable. In some instances, IgA deposition causes no changes in glomerular morphology, whereas severe inflammation including crescent formation occurs in others. In vitro, IgA immune complexes induce mesangial cell proliferation and secretion of extracellular matrix components⁹⁷ (Fig. 2). Three mesangial cell receptors for IgA have been described: transferrin receptor (CD71), integrins $\alpha 1 \beta 1$ and $\alpha 2 \beta 2$, and β -1,4-galactosyltransferase 1 (ref. 98). Among the pathways activated by IgA immune complexes, spleen tyrosine kinase (SYK) signalling is of particular interest, as it is activated in mesangial cells in

IgAN but also implicated in adaptive and innate immune responses^{99,100}. Furthermore, inhibition of SYK signalling in vitro suppresses the inflammatory response of human mesangial cells exposed to IgA1 from patients with IgAN¹⁰¹. Another pathway is the MAPK–ERK signalling pathway, which is activated in the mesangium of patients with IgAN, proteinuria >1 g per day and elevated blood pressure. Moreover, IgA1-dependent ERK activation depends on angiotensin II type 1 receptor activation, as its inhibition effectively reduces proteinuria in patients with pronounced mesangial ERK activation. Assessment of MAPK–ERK activation in diagnostic renal biopsies was suggested to predict the therapeutic efficacy of renin–angiotensin–aldosterone system inhibition (RAASi) in IgAN¹⁰².

Immune complexes in IgAN also have an enhanced affinity for glomerular endothelial cells and induce glomerular filtration barrier dysfunction, enhancing subsequent inflammatory responses¹⁰³. Soluble mediators derived from mesangial cells alter podocyte gene expression and glomerular permeability, leading to podocyte damage and segmental glomerulosclerosis^{48,104–106}. Additionally, these mediators can activate proximal tubular epithelial cells driving tubulointerstitial fibrosis^{107,108} (Fig. 2).

The complement system. Mesangial complement deposition is common in IgAN (Fig. 3) and the extent of mesangial C3 deposition correlates with the severity and progression of IgAN^{109,110}. Patients with IgAN rarely have reduced serum C3 levels outside the normal range, however reduced serum C3 levels have been associated with a poor prognosis^{111–113}. In addition to C3, deposits of complement factor H (CFH), mannose-binding lectin (MBL), membrane attack complex (MAC), C4d and properdin are variably detected in the mesangium in IgAN, but C1q deposits are usually absent^{114–119}. However, in a case series from China, 14% of kidney biopsy samples contained C1q deposits, which was an independent risk factor for kidney failure¹²⁰.

Polymeric IgA, aggregates of IgA and Gd-IgA1 can all activate the alternative complement pathway^{121–123}. Alternative pathway complement proteins, including factor B, CFH and the factor H-related proteins (FHRs) encoded by CFHR, can be detected in glomeruli in IgAN¹²⁴. In addition, two separate GWA studies have linked the CFH/CFHR locus with IgAN risk, finding that deletion of complement FHR1 and FHR3 protects against the development of IgAN^{20,21}. Similarly, FHR5 deposition was associated with more severe histological injury in a cohort of Chinese patients with IgAN¹²⁵. CFH has also been detected in the urine in IgAN and positively associated with disease progression¹²⁶. However, increased FHR1 levels have also been found in patients with autosomal dominant polycystic kidney disease and kidney function deterioration, suggesting that kidney function impairment non-specifically elevates FHR1 levels¹²⁷.

Glomerular deposition of components of the lectin pathway occurs in a subset of patients with IgAN and is associated with disease severity. In vitro evidence shows that polymeric IgA can activate the lectin pathway^{128,129}. The presence of mesangial C4d with absent C1q, which is indicative of lectin pathway activation, is associated with more rapid progression to kidney failure in IgAN¹²⁹. C4d deposits are not restricted to glomeruli in IgAN but are also found in the kidney vasculature, particularly in arterioles, which is also associated with faster disease progression¹³⁰. Urine levels of MBL and C4d correlate with their respective mesangial deposits and may mirror disease activity in IgAN¹³¹.

Glomerular deposition of C5b9 (MAC) in IgAN is common and correlates with the presence of mesangial C3b/iC3b/C3c and C3d^{115,132,133}.

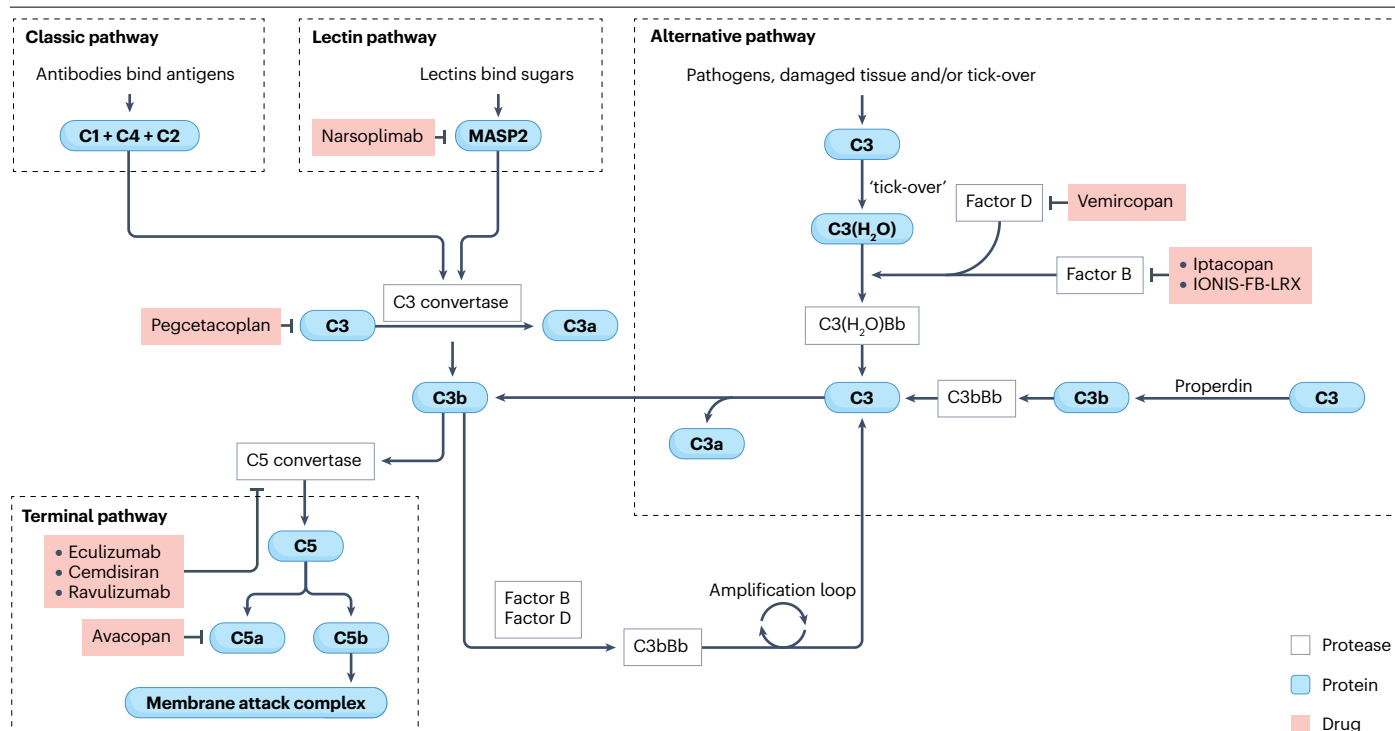


Fig. 3 | The complement system and targeted drug treatments. The complement cascade can be activated through different pathways: the classic pathway, the lectin pathway and the alternative pathway. Regardless of the trigger, the key step in complement activity is the activation of C3. In IgA nephropathy (IgAN), the alternative and lectin pathways are particularly relevant in the pathophysiology of the disease. The lectin pathway is triggered by complexes of pattern recognition molecules (PRMs, such as mannose-binding lectins (MBL), collectins and ficolins) and serine proteases, in which PRMs bind to carbohydrate motifs on microbial surfaces. Mannose-binding-lectin-associated serine protease 1 (MASP1) and MASP2, become activated and cleave C4 and C2, similar to the classic pathway, and result in the formation of C4bC2b, the C3 convertase of the classic and lectin pathways. By contrast, the alternative pathway continuously self-activates at a low level ('tick-over'), leading to the generation of C3b, the key effector molecule of the complement system, which deposits on pathogens, immune complexes or debris. Activation of the

alternative complement pathway requires cleavage of factor B by factor D into Ba and Bb, which bind to C3b to form C3bBb, and the alternative pathway C3 convertase that is further stabilized by properdin to enhance C3 activation. Thus, all three pathways converge at the formation of C3 convertases, which convert C3 into C3a (an anaphylatoxin) and C3b. The binding of another C3b to the C3 convertase generates the C5 convertase, which then cleaves C5 to release the potent anaphylatoxin C5a. Additionally, C5b initiates the sequential formation of the lytic membrane attack complex as part of the terminal pathway that causes cell and tissue damage as well as inflammation. Current and new drugs are being developed to target specific components of the lectin and alternative pathways aiming to modulate their activation and control inflammation in IgAN. Other conditions that are dependent on complement activation, such as atypical haemolytic uraemic syndrome and paroxysmal nocturnal haemoglobinuria, are also being treated with inhibitors of complement.

C5b-9 can also be found in the capillary wall, Bowman's capsule, tubular basement membrane and vascular wall¹³⁴. Glomerular C5b-9 deposition in IgAN correlates with the extent of glomerulosclerosis, mesangial expansion, hypercellularity, interstitial inflammation and fibrosis, whereas tubular C5b-9 deposition is linked to tubular atrophy, interstitial inflammation and interstitial fibrosis¹³⁴. Both glomerular and tubular deposition of C5b-9 is associated with kidney function, proteinuria and progressive IgAN¹³⁴.

Of note, despite this plethora of descriptive studies, no non-invasive biomarkers, including complement component levels in blood or urine, have emerged to reliably assess disease activity or intrarenal complement activation in patients with IgAN.

Diagnosis, screening and prevention

IgAN can only be diagnosed by kidney biopsy. Although levels of IgA, IgA1, specific IgA1 O-glycoforms and IgA immune complexes may be elevated in the serum, their use as diagnostic tests is not yet established¹³⁵.

Episodic visible haematuria or incidentally detected asymptomatic non-visible haematuria should be taken seriously and patients should be advised to see a nephrologist for further work-up and potential kidney biopsy. Varying levels of proteinuria, ranging from mild (<1 g per day) to severe (>3 g per day), might be present in parallel to haematuria.

Globally, scheduled urinary screening for IgAN only exists in Japan and South Korea, where the prevalence and incidence of glomerulonephritis, particularly IgAN, are notably higher than in other regions^{136,137}. Nonetheless, early diagnosis is beneficial for the patients as early intervention can prevent kidney function decline. Patients with chronic liver disease, inflammatory bowel disease and, in particular, IgAV should receive urinary screening during check-up visits to detect signs of concomitant IgAN early. Conversely, patients with diagnosed IgAN should be evaluated for signs of secondary causes. To date, there is no information on prevention, as IgAN is believed to arise from a primary biochemical defect in the mucosal IgA1 molecule itself according to the multi-hit hypothesis as alluded to above.

Box 2

Childhood IgAN

IgA nephropathy (IgAN) in children (cIgAN) generally follows a milder course than IgAN in adults. In most instances, children with IgAN experience episodes of gross haematuria that spontaneously resolve or persist as microscopic haematuria. The long-term prognosis of cIgAN is typically favourable, with a 10-year survival rate of ~91% and 20-year survival rate of 81%^{289–291}. However, it is essential to not underestimate the risk of disease progression, especially considering the long life expectancy of young patients. Interestingly, a considerable number of individuals who eventually require kidney replacement therapy for IgAN are young adults, aged between 24 and 54 years, with 22% of them being <30 years old²⁹². This observation suggests that the disease may have its origins in the paediatric age and evolve over decades.

The pathogenetic events leading to IgAN development are largely similar in both children and adults²⁹³. Paediatric IgAN also activates mucosal immunity, with upper respiratory tract infections being more common in children, implying potential differences in the initial triggers between age groups. However, a persistently activated gut-associated lymphoid tissue is a shared feature among children and adults with IgAN, suggesting common aetiological events with potentially distinct triggers²⁹³.

Genetic susceptibility plays a role in the age at IgAN onset, showing an association between genetic risk scores and the age at diagnosis. This suggests a shared genetic background in paediatric and adult IgAN, with common risk alleles favouring disease development in children²⁹³. Clinical risk factors for progression of IgAN differ between children and adults. Parameters such as proteinuria and eGFR at kidney biopsy are less predictive in children²⁹³. Conversely, blood pressure at kidney biopsy is a stronger risk factor, often linked to irreversible sclerotic kidney damage, while younger age consistently indicates better outcomes²⁹³.

There is no evidence opposing the use of different treatments in children and adults with IgAN. Treatment with corticosteroid or immunosuppressive agents in cIgAN (and paediatric IgAV) is common in Asia²⁹⁴ and in Europe²⁹⁵. In both groups, assessing the risk profile is crucial to avoid overtreatment in benign disease and initiate timely and appropriate therapy in potentially progressive disease.

Clinical presentation

IgAN manifestations range from asymptomatic non-visible or visible haematuria with stable kidney function to acute kidney injury (AKI) with rapidly progressive glomerulonephritis (RPGN)¹. In children and young adults, around 30–40% of patients present with episodic visible haematuria in conjunction with an upper respiratory tract infection or, less commonly, with gastrointestinal tract infections (Box 2). Haematuria typically occurs 1–2 days after the onset of infectious symptoms. Around 40–50% of patients present with asymptomatic non-visible haematuria with varying degrees of proteinuria (usually <2 g per day), with or without progressive kidney failure. These signs are often detected incidentally through routine urine analysis. Less than 5% of patients develop clinically relevant AKI, but AKI may be the initial presentation

in up to 27% of patients >65 years of age¹³⁸. The two mechanisms that can cause AKI are acute glomerular injury with RPGN and, far more commonly, acute tubular injury from red blood cell cast obstruction and/or haem toxicity during episodes of visible haematuria. The latter is usually reversible, and a lack of improvement after 1 to 2 weeks indicates an alternative pathology and the need for a kidney biopsy¹. Risk factors for incomplete recovery of kidney function after AKI include a duration of visible haematuria of >10 days, being >50 years old, a decreased baseline eGFR, the absence of previous episodes of visible haematuria and the severity of tubular necrosis¹³⁹. Nephrotic syndrome is an infrequent presentation, occurring in only 5% of patients. A kidney biopsy in these patients may show mesangial deposition of IgA and extensive foot process effacement, suggesting minimal change disease with coincidental IgA deposits. Conversely, proteinuria in the nephrotic range without hypoalbuminaemia may occur in the setting of advanced secondary focal segmental glomerulosclerosis.

Secondary IgA nephropathy. IgAN has been documented in patients with various co-morbidities, but it is not clear whether these secondary forms share common pathophysiological pathways. Abnormally glycosylated IgA1 and IgA–IgG complexes have been detected in secondary forms¹⁴⁰. There is no consistent definition of secondary IgAN in the literature and there are no histological features that distinguish it from primary IgAN¹⁴¹. Given that mesangial IgA deposits have been observed in 4–16% of healthy individuals, some reported associations may be coincidental rather than causal^{142,143}.

Among the secondary IgAN forms, IgAV and IgAN associated with chronic liver disease, particularly alcoholic cirrhosis, are the most common diagnoses^{141,144,145}. IgAV-associated nephritis and IgAN may have a similar pathogenesis as the histological findings in the kidney are identical, and high levels of Gd-IgA1 have been found in patients with either condition^{146–148}. Some authors have suggested that they are two manifestations of the same disease, but they differ in presentation and age at onset¹⁴⁹.

Patients with IgAN have an increased risk of inflammatory bowel disease both before and after their nephropathy diagnosis¹⁵⁰. Clinical courses of both diseases closely correlate, and treatment of the bowel disease is usually associated with clinical remission and biopsy-proven clearance of mesangial IgA deposits¹⁵⁰. In addition, among patients with IgAN, comorbid inflammatory bowel disease was found to elevate the risk of progression to kidney failure⁵⁹. Coeliac disease has also been associated with IgAN¹⁵¹. In a population-based prospective cohort study, patients with coeliac disease had a threefold increased risk of developing IgAN in the future¹⁵².

Infections have also been associated with the development of IgAN, but many of these associations may be coincidental rather than causal^{141,153}. IgA-dominant infection-associated glomerulonephritis (IAGN) is an increasingly recognized entity. IAGN is most commonly secondary to staphylococcal infections, with risk factors for an unfavourable prognosis including age and diabetes mellitus¹⁵⁴. Based on differences in clinicopathological findings, the pathophysiology of IAGN is thought to be distinct from that of primary IgAN^{141,154,155}.

Finally, IgAN cases have been described in association with several autoimmune disorders, including psoriasis and ankylosing spondylitis, and with malignant processes^{156–158}.

Pathology

The diagnosis of IgAN requires a kidney biopsy showing dominant or co-dominant mesangial IgA staining.

Light microscopy. The light microscopic appearance of IgAN varies considerably between patients and commonly reflects prior disease activity and the clinical course. Biopsy samples can appear almost normal but with immunohistochemically detectable IgA deposits, they can show acute inflammatory lesions (including mesangial hypercellularity, endocapillary proliferation and crescent formation) or they can comprise chronic scarring lesions (segmental and global glomerulosclerosis and tubulointerstitial fibrosis). As kidney function declines secondarily, adaptive focal and segmental glomerulosclerosis becomes more common. Mesangial proliferation and matrix expansion are the most common morphological features of IgAN (Fig. 4). Isolated cellular crescents occur frequently, especially in association with poorly controlled blood pressure. In cases of rapidly progressive IgAN, defined as a 50% loss of eGFR over 3 months, extensive crescent formation associated with focal necrotizing lesions is commonly seen. The extent of tubulointerstitial fibrosis and tubular atrophy is strongly correlated with eGFR and the extent of chronic kidney damage, reflecting prior disease activity and indicating prognosis.

Immunohistology and electron microscopy. The hallmark of IgAN is dominant or co-dominant mesangial IgA deposits, which are visualized by immunohistochemistry and are the sole diagnostic criterion of IgAN (Fig. 4). These deposits are often accompanied by C3 co-deposition and, infrequently, deposition of IgG and IgM. Deposits can be found both in the mesangium and less commonly in the capillary loops. Accordingly, mesangial and paramesangial electron-dense deposits can be detected by electron microscopy (Fig. 4).

Prognostic biomarkers

Biomarkers that can predict an individual's disease course would be of great benefit. To date, the prognosis of IgAN largely depends on the extent of proteinuria, eGFR and the control of blood pressure^{159–161}. Thus, frequent clinical monitoring over decades is useful even in patients with low disease activity, as worsening of kidney function can

occur several years after initial diagnosis. CKD progression has been described within 22 years in 20% of patients with 'benign' IgAN of whom 3% subsequently developed kidney failure¹⁶². Of note, in some patients traditionally categorized as having low-risk disease based on proteinuria levels of <0.88 g/g, the incidence of kidney failure within just 10 years of diagnosis was alarmingly high, underscoring the importance of continuous vigilance and monitoring¹⁶³.

Initial histopathological findings are important to assess prognosis. For this purpose, IgAN biopsy samples are scored according to the Oxford MEST-C classification^{164,165}. Five parameters are recorded in MEST-C, namely mesangial hypercellularity (M), endocapillary proliferation (E), segmental glomerulosclerosis (S), tubular atrophy and interstitial fibrosis (T) and glomerular crescents (C) (Table 2). A large European validation study in more than 1,100 patients with IgAN found that, during a median follow-up of 7 years, in particular M1, S1 and T1–T2 lesions were predictors of poorer kidney survival¹⁶⁶. The Oxford MEST-C classification is widely used, but its ability to predict the need for glucocorticoid therapy remains uncertain due to inconsistent findings. Accordingly, the 2021 KDIGO Guideline states that there is insufficient evidence to support its use to determine when any glucocorticoid therapy should be commenced⁵⁴. In 2022, a new scoring system was introduced that combines active and chronic lesions. This approach revealed a benefit from immunosuppression in reducing the risk of kidney failure in individuals with highly active and mildly chronic pathological lesions, in contrast to those with chronic lesions¹⁶⁷. Although this scoring system was validated in the European VALIGA cohort¹⁶⁸, it is essential to stress the need for prospective randomized controlled trials that evaluate whether therapeutic decisions can indeed be based on 'snap-shot' histological findings.

Additionally, an international IgAN prediction tool has been developed to predict prognosis at the time of biopsy. The original tool has been revised and can be used up to 2 years after kidney biopsy to evaluate the risk of a 50% decline in kidney function or kidney failure within 80 months (6.7 years). A prediction tool for children has also been

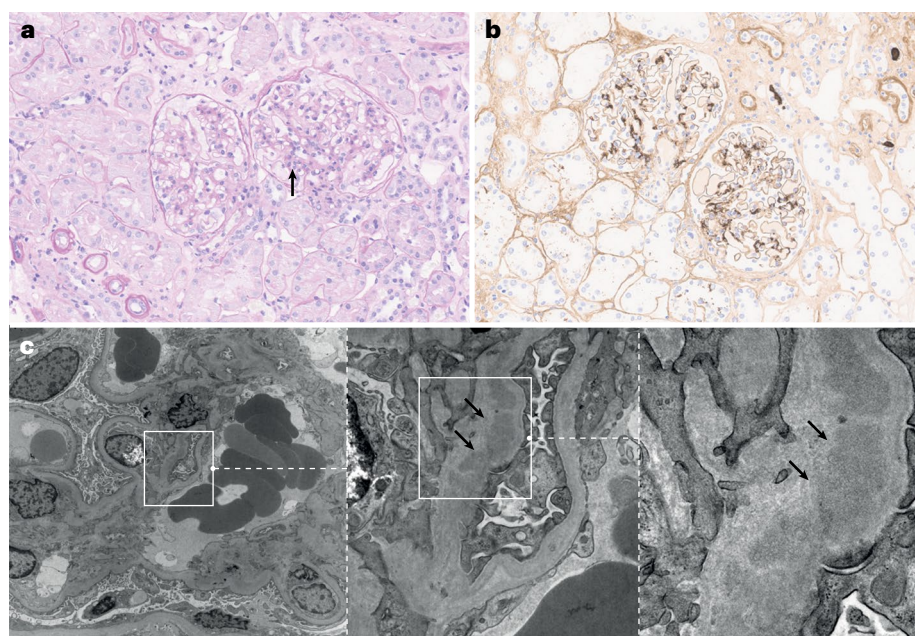


Fig. 4 | Renal biopsy findings in IgAN. **a**, Light microscopy image showing kidney cortex with two glomeruli (in the centre) surrounded by tubulointerstitium. The glomeruli show an increase in the mesangial cells (arrow), termed mesangial hypercellularity and representing the M criterion of the MEST-C classification. Periodic acid Schiff (PAS) staining (original magnification of whole slide image, $\times 20$). **b**, Immunohistochemical staining for IgA in two glomeruli (in the centre) in a section deeper than panel **a**. The dark brown colour represents the typical pattern of IgA deposition in the mesangium (original magnification of whole slide image, $\times 20$). **c**, Transmission electron microscopy images showing glomerular capillary loops and the mesangium with electron-dense mesangial and paramesangial deposits (arrows). The centre and right images are digital magnifications of the left image ($\times 2,784$, $\times 10,000$ and $\times 27,800$, respectively). IgAN, IgA nephropathy.

Table 2 | MEST-C Score Oxford classification of IgAN¹⁶⁴

Histological finding	Definition	Score
Mesangial hypercellularity (M)	More than four mesangial cells in any mesangial area of a glomerulus	M0: $\leq 50\%$ of glomeruli showing mesangial hypercellularity M1: $>50\%$ of glomeruli showing mesangial hypercellularity
Endocapillary hypercellularity (E)	Hypercellularity due to an increased number of cells within the glomerular capillary lumina	E0: absent E1: present in any glomerulus
Segmental glomerulosclerosis (S)	Adhesion or sclerosis (obliteration of capillary lumina by matrix) in part but not the whole glomerular tuft	S0: absent S1: present in any glomerulus
Tubular atrophy/interstitial fibrosis (T)	Estimated percentage of cortical area showing tubular atrophy or interstitial fibrosis, whichever is greater	T0: 0–25% of the cortical area T1: 26–50% of the cortical area T2: $>50\%$ of the cortical area
Crescents (C)	Percentage of glomeruli with cellular or fibrocellular crescents	C0: absent C1: 0–25% of glomeruli C2: $\geq 25\%$ of glomeruli

IgAN, IgA nephropathy.

created¹⁶⁹. In addition to the clinical parameters eGFR, blood pressure, proteinuria and age, information on treatment (RAASi and immunosuppression), race and the variables of the MEST-C classification are included in the formula^{170,171}. The calculator has been validated in several studies and is also recommended by the 2021 KDIGO Guideline to be used in counselling individual patients¹⁷². It has not been developed to guide treatment and should not be used for this purpose today.

Management

Supportive care

All patients with IgAN should receive optimized supportive care according to the 2021 KDIGO Guideline⁵⁴ (Fig. 5). The importance of comprehensive supportive care in IgAN is highlighted by the results of the STOP-IgAN study, in which almost one-third of participants responded well to intensive supportive care and were no longer eligible for randomization¹⁷³. Optimal supportive care includes blood pressure control, maximally tolerated RAASi, treatment of dyslipidaemia and lifestyle changes (low-sodium diet defined as <2.0 g (90 mmol) per day, weight reduction and smoking cessation). RAASi is the first-line treatment in patients with IgAN, with proven effectiveness in reducing proteinuria and slowing disease progression^{174,175} (Fig. 2). Although no controlled clinical trials have investigated whether RAASi is beneficial in normotensive patients with IgAN and only moderately increased proteinuria (that is, ≤ 0.5 g per day), they should also be candidates for optimized supportive care. Full RAASi by adding a direct renin inhibitor in addition to an

angiotensin II receptor blocker did not further improve outcomes and may induce more adverse effects¹⁷⁶. Dihydropyridine calcium-channel blockers should be avoided as first-line agents, as they induce pre-glomerular vasodilation, transmitting high blood pressure to the glomerulus^{177–181}. High BMI and nicotine consumption are independent risk factors for IgAN progression and should be addressed^{182–184}.

The use of SGLT2 inhibitors has emerged as a promising new approach in IgAN¹⁸⁵. It is expected that, when the KDIGO Guideline is updated, SGLT2 inhibitors will have a prominent role in the generic management of all patients with proteinuric kidney diseases (Fig. 2). Moreover, new research has identified endothelin 1 as a promising therapeutic target. Sparsentan, a dual endothelin receptor and angiotensin receptor antagonist, has received FDA accelerated approval owing to its efficacy in reducing proteinuria in the PROTECT trial¹² (Fig. 2). Meanwhile, atrasentan, a selective endothelin A receptor inhibitor, is also being evaluated in the ongoing phase III ALIGN study (NCT04573478)¹⁸⁶. Following positive results in patients with CKD and type 2 diabetes¹⁸⁷, finerenone is being evaluated in the FIND-CKD trial (NCT05047263) to assess its efficacy in delaying CKD progression in non-diabetic patients¹⁸⁸. If successful, finerenone could become another valuable addition to our armamentarium of supportive care options.

Immunosuppression

Corticosteroids. The 2021 KDIGO Clinical Practice Guideline⁵⁴ suggests considering a 6-month course of systemic corticosteroid therapy in patients with IgAN and proteinuria of >1 g per day, despite maximal supportive care for 3–6 months, and an eGFR of >50 ml/min per 1.73 m², after discussing the risk of treatment-related toxic effects with patients⁵⁴ (Fig. 5). The safety and efficacy of this treatment option are still controversial. The STOP-IgAN trial found no difference in annual decline in eGFR over a median follow-up of 7.4 years between the supportive care group and the immunosuppression group^{173,189}. Interestingly, even though the primary outcome of full clinical remission was more often achieved in the immunosuppression group, it did not translate into long-term beneficial effects on kidney outcomes. Moreover, the corticosteroid group experienced a higher incidence of adverse events, particularly infections. Meanwhile, the TESTING trial confirmed the concerns about the safety of corticosteroids, resulting in early termination. The follow-up study of the TESTING trial supported the renoprotective effects of both full-dose and reduced-dose oral corticosteroid therapy on both the eGFR and proteinuria after a mean follow-up of 4.2 years¹⁹⁰. Although the improved safety profile observed in the second part of the study, in which concomitant prophylactic co-trimoxazole was used, was notable, it is important to point out that despite these measures one infection-related death still occurred in the steroid group. Although both trials acknowledged the increased risk of infectious adverse events associated with corticosteroid use, they arrived at opposing conclusions regarding the effects of corticosteroids on kidney outcomes. Although proteinuria decreased with the use of corticosteroids in both studies, the effect was transient in STOP-IgAN. The different outcomes may have been due to differences in the patient populations studied. The STOP-IgAN trial only included white German patients, whereas 95% of participants in the TESTING trial were of Asian ancestry, which is a population known to have more aggressive disease, higher baseline proteinuria levels and a more rapid decline in kidney function¹⁹¹.

The rationale for using immunomodulatory therapy in IgAN is supported by its presumed immune-mediated pathogenesis but, owing to its chronic progressive nature, the potential need for longer or repeated

courses of systemic corticosteroids and their adverse effects makes this approach problematic^{192,193}. To reduce treatment-associated adverse effects, TRF-budesonide, which releases the drug specifically in the distal ileum where there is a high density of mucosal Peyer's plaques and B lymphocytes, has been developed¹⁹⁴ (Fig. 2). The therapeutic potential of TRF-budesonide was tested in the NeflgArd study, an international multicentre double-blind phase III study¹⁹⁵. At 9 months, proteinuria (urine protein-to-creatinine ratio) was 27% lower in the TRF-budesonide group than in the placebo group and continued to fall in the months after treatment cessation, along with a benefit in eGFR preservation. TRF-budesonide was well tolerated, and treatment-emergent adverse events were mostly mild to moderate in severity and reversible. In particular, no infectious complications (such as acne), weight gain and diabetes occurred. Published 2-year follow-up data demonstrated a durable reduction in proteinuria and a significant GFR protection¹⁰. TRF-budesonide was granted accelerated approval by the FDA and conditional market authorization by the EMA, and it is the first approved treatment of IgAN. Non-randomized small off-label studies have also explored the therapeutic potential of classic budesonide in

patients with IgAN¹⁹⁶ but no head-to-head comparison with the targeted release formulation is currently available. Another small short-term study limited to Chinese patients suggested inhalation with fluticasone propionate as an alternative therapeutic approach¹⁹⁷. However, further investigation is needed.

Other immunosuppressive agents. Apart from glucocorticoids, the 2021 KDIGO Guideline⁵⁴ does not recommend any other immunosuppressive therapies in IgAN, such as azathioprine, cyclophosphamide and calcineurin inhibitors. However, mycophenolate mofetil (MMF) can be considered as an alternative to steroids or as a steroid-sparing agent in Chinese patients, as adverse events associated with MMF are lower than those observed with systemic glucocorticoids in MMF studies^{198–202}. In an open-label trial in 170 Chinese patients with IgAN and persistent proteinuria, those randomized to MMF monotherapy plus supportive care had a reduced risk of disease progression compared with those receiving supportive care alone²⁰³. In contrast to these Chinese studies, both a Belgian trial and a North American trial that recruited patients with more advanced disease did not note any

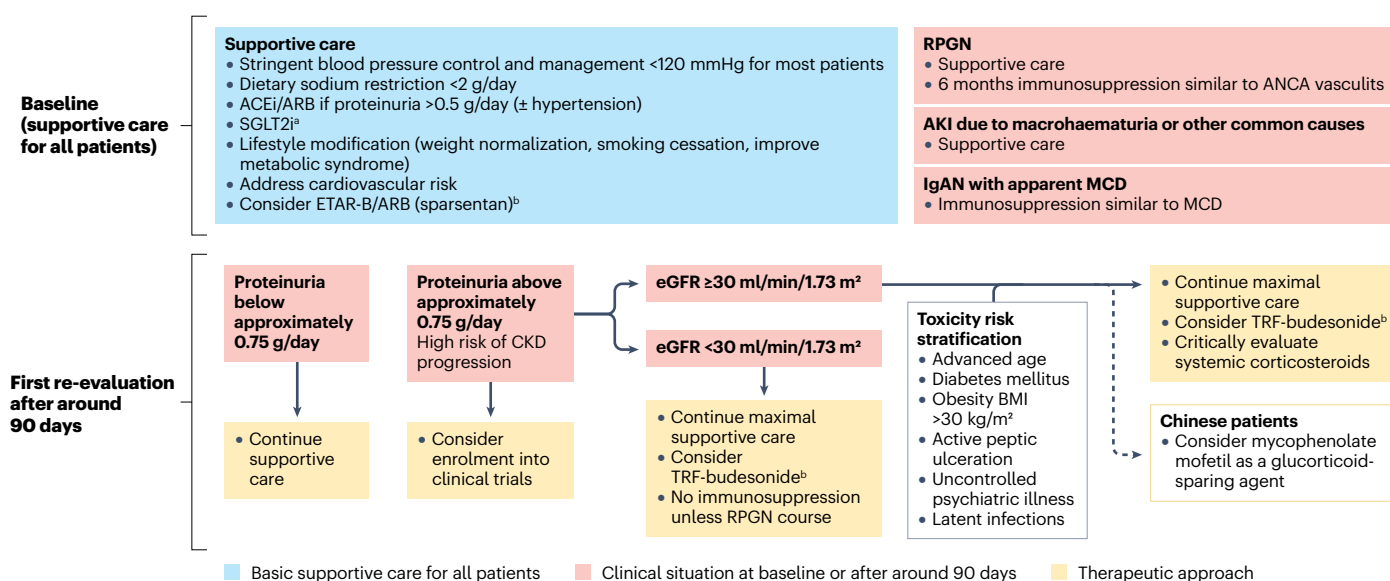


Fig. 5 | Modified KDIGO 2021 therapeutic algorithm. In the first step (blue box), all patients diagnosed with IgA nephropathy (IgAN) should initially receive an optimization of supportive care including an sodium–glucose co-transporter 2 inhibitor (SGLT2i)^a and possibly sparsentan^b. In step two after around 90 days, re-evaluation is necessary (bottom red boxes). Patients maintaining proteinuria above approximately 0.75 g/day and with an estimated glomerular filtration rate (eGFR) >30 ml/min should be considered for corticosteroid treatment either systemically, following a thorough risk assessment for corticosteroid-related toxic effects, or for therapy with targeted release formulation budesonide (TRF)-budesonide^b. In addition, enrolment in clinical studies should be considered. Patients with proteinuria above approximately 0.75 g/day and an eGFR of <30 ml/min should continue on maximal supportive therapy, with the option of considering therapy with targeted release formulation budesonide (TRF-budesonide)^b. The NeflgArd trial^{10,195} assessed TRF-budesonide^b effectiveness in patients with an eGFR of >30 ml/min. Although its use in patients with an eGFR of <30 ml/min has not been tested, it should be considered owing to its lower risk of adverse effects than systemic corticosteroid therapy. Mycophenolate mofetil should be considered as a glucocorticoid-sparing agent in Chinese patients, but there is currently no evidence to recommend its use in

non-Chinese patients. Major changes of this approach compared with the 2021 KDIGO Guideline include the incorporation of SGLT2i, TRF-budesonide and sparsentan¹², all of which were approved for use in patients with chronic kidney disease (CKD) (SGLT2i) or IgAN (TRF-budesonide, sparsentan) after publication of the KDIGO Guideline. Future studies will have to address whether early immunomodulatory therapy initiated right after kidney biopsy in patients with high inflammatory activity is therapeutically more effective than a 90-day run-in period with optimization of supportive care. Special situations include (red boxes on the right) patients diagnosed with rapidly progressive glomerulonephritis (RPGN), who should receive immunosuppression and corticosteroids. In those with nephrotic syndrome and presence of global foot process effacement on electron microscopy, treatment strategies should align with those for patients with minimal change disease (MCD). AKI, acute kidney injury; ARB, angiotensin II receptor blocker; ETAR-B, endothelin A receptor blocker. ^aIn most regions of the world, SGLT2i are approved for use in non-diabetic CKD as long as eGFR is >25 ml/min; however, SGLT2i should not be stopped if the eGFR continues to fall to <25 ml/min. ^bBoth sparsentan and TRF-budesonide have, to date, received FDA and/or EMA conditional approval to reduce proteinuria in patients with high-risk IgAN (defined as urine protein-to-creatinine ratio of ≥1.5 g/g).

benefit from MMF^{198,199}. Thus, the 2021 KDIGO Guideline limits its MMF recommendation to patients with IgAN of Chinese origin.

Hydroxychloroquine, a potent TLR9 and, to a lesser extent, TLR8 and TLR7 inhibitor, has been evaluated for the treatment of IgAN in a single-centre Chinese study (Fig. 2). Hydroxychloroquine in addition to optimized RAASi reduced proteinuria over 6 months without evidence of adverse events²⁰⁴. Long-term data indicate that hydroxychloroquine with conventional RAASi can reduce proteinuria by ~50% within 24 months without significant decline in eGFR and serious adverse events, highlighting hydroxychloroquine as a potentially effective and safe supportive treatment for Chinese patients in the long term²⁰⁵.

Novel agents and ongoing clinical trials. Targeting B cells, which produce Gd-IgA1 and anti-Gd-IgA1 antibodies, is an attractive therapeutic approach. Rituximab, a chimeric monoclonal antibody against CD20, may seem to be an ideal candidate, but a small open-label trial in IgAN failed to demonstrate benefit²⁰⁶. Despite effectively reducing circulating B cells, plasma cell counts remained unaffected and levels of serum Gd-IgA1 and anti-Gd-IgA1 antibodies were not reduced. This finding led to the hypothesis that CD20-negative plasma cells may have a crucial role in the production of Gd-IgA1. A pilot study using bortezomib, a proteasome inhibitor, provided support for this approach by demonstrating a reduction in proteinuria in eight patients with IgAN, of whom three experienced full disease remission. Currently, a phase IIa clinical trial is underway in patients with IgAN to investigate the use of felzartamab, a fully human IgG1 monoclonal antibody that targets CD38⁺ cells (NCT05065970)²⁰⁷ (Table 3 and Fig. 2).

Other approaches targeting B cells involve inhibiting BAFF and APRIL using specific monoclonal antibodies or drugs (Table 3 and Fig. 2). Dual inhibitors atacicept and telitacept are currently being tested in clinical trials in patients with IgAN (NCT05799287 and NCT04716231)^{208,209}. Data from phase II studies of both agents suggest that this approach affects the production of Gd-IgA1, resulting in a reduction in proteinuria for up to 69 months^{210,211}. There are also

first data on sibeprenlimab (VIS649) (NCT04287985)²¹² and zigakibart (BION-1301) (NCT03945318)²¹³, both of which specifically target APRIL, are well tolerated and reduce proteinuria in patients with IgAN^{211,214}. Blisibimod, a monoclonal antibody against BAFF, was evaluated in the phase II/III study BRIGHT-SC (NCT02062684)²¹⁵ but the study data have not been published even though the trial phase ended in 2016. A phase II clinical study (NCT02112838) evaluating the safety and efficacy of fostamatinib, a SYK inhibitor, in IgAN has been completed²¹⁶. Although there was a dose-dependent reduction in proteinuria, it did not reach statistical significance²¹⁷.

Several ongoing phase II/III trials are evaluating complement-directed therapies (Table 3 and Fig. 3). Initial case reports of C5 inhibition with eculizumab in rapidly progressive IgAN have provided proof of concept for further evaluation of complement inhibition in IgAN. Ravulizumab (ALXN1210) is a long-acting C5-blocking antibody currently being evaluated in a phase II clinical trial (NCT04564339)²¹⁸. Cemdisiran (ALN-CC5), an investigational RNAi therapeutic that inhibits hepatic production of C5, has shown promising results in a phase II study (NCT03841448). Monthly subcutaneous doses of cemdisiran led to a reduction in proteinuria at week 32 relative to placebo, with 31.8% of patients achieving >50% reduction²¹⁹.

Iptacopan (LNP023) is a new oral highly selective inhibitor of factor B of the alternative pathway. In a phase II study (NCT03373461), iptacopan was well tolerated and led to continuous reductions in proteinuria and strong inhibition of alternative pathway activity over 6 months^{220,221}. The phase III APPLAUSE-IgAN trial with iptacopan is ongoing (NCT04578834)²²². Pegcetacoplan (APL-2) is another complement inhibitor that targets C3 and is currently being evaluated in a phase II study (NCT03453619)²²³. IONIS-FB-LRx (RG6299) is a ligand-conjugated antisense inhibitor of complement factor B and is also being tested for safety and efficacy in patients with IgAN (NCT04014335)²²⁴. Narsoplimab (OMS721) is a fully human IgG4 monoclonal antibody that inhibits MBL-associated serine protease 2 (MASP2), the key effector enzyme of the lectin pathway. In a phase II clinical trial

Table 3 | Therapeutic agents currently in clinical trials for the treatment of IgAN

Drug	Target	Phase	Route of administration	NCT	Ref.
Felzartamab	Humanized IgG1 monoclonal antibody against CD38	IIa	Intravenous	NCT05065970	207
Fostamatinib	SYK inhibitor	II	Oral	NCT02112838	216
Sibeprenlimab	Humanized IgG2 against APRIL	II	Intravenous	NCT04287985	212
Zigakibart	Humanized IgG4 against APRIL	I/II	Intravenous	NCT03945318	213
Blisibimod	Selective inhibitor of BAFF	II/III	Subcutaneous	NCT02062684	215
Telitacept	Dual inhibitor of BAFF/APRIL	III	Subcutaneous	NCT05799287	208
Atacicept	Antagonist of BAFF/APRIL	II/III	Subcutaneous	NCT04716231	209
Iptacopan	CFB inhibitor	III	Oral	NCT04578834	222
IONIS-FB-LRx	Antisense inhibitor of CFB	II	Subcutaneous	NCT04014335	224
Pegcetacoplan	Complement C3 inhibitor	II	Subcutaneous	NCT03453619	223
Narsoplimab	Humanized MASP2 monoclonal antibody	III	Intravenous	NCT03608033	227
Ravulizumab	Humanized anti-C5 monoclonal antibody	II	Intravenous	NCT04564339	218
Cemdisiran	RNAi targeting complement C5	II	Subcutaneous	NCT03841448	171
Vemircopan	Factor D inhibitor	II	Oral	NCT05097989	284

APRIL, a proliferation-inducing ligand; BAFF, B cell activation factor; CFB, complement factor B; IgAN, IgA nephropathy; MASP2, mannose-binding-lectin-associated serine protease 2.

Glossary

Alternative complement pathway

The evolutionarily oldest of the three complement pathways. It undergoes constant low-level self-activation, which is rapidly amplified in the presence of a microorganism, a damaged host cell, or a deficiency in complement regulatory proteins.

BAFF

A key cytokine and member of the TNF family of ligands that regulates B cell survival, maturation and differentiation through binding with its receptors: BAFF receptor (BAFF-R), transmembrane activator and calcium-modulator and cyclophilin ligand interactor (TACI), and B cell maturation antigen (BCMA).

Class switch recombination

A process by which a mature B cell switches from the production of IgM

to the production of IgG, IgE or IgA in response to antigen stimulation and costimulatory signals.

Complement C3

A pivotal protein in the activation of the complement system, with its processing and activation by C3 convertase being essential for both the classic and alternative pathways of complement activation.

Crescent

Inflammatory collection of cells in Bowman's space resulting from severe glomerular injury and localized rupture of the capillary wall or Bowman's capsule, which enables entry of plasma proteins and inflammatory material into Bowman's space.

eGFR

The estimated overall rate at which fluid is filtered through the kidney;

it is commonly measured using renal clearance techniques and typically averages around 120 ml/min per 1.73 m² of surface area.

Glomerulosclerosis

Scarring of glomeruli that disrupts the filtration process and leads to leakage of protein from the blood into the urine.

Lectin pathway

One of the three complement pathways. It is activated by pattern recognition molecules of two major protein families, the ficolins and the collectins. This pathway has a crucial role in innate immunity and host defence against microbial infections.

Minimal change disease

A podocytopathy that is characterized by the absence of histological abnormalities except for ultrastructural evidence of podocyte foot process

fusion and, clinically, by the manifestation of nephrotic syndrome.

Nephrotic syndrome

A clinical syndrome characterized by the presence of heavy proteinuria (>3.5 g per 24 h), hypoalbuminaemia, hypercholesterolaemia and peripheral oedema due to increased filtration of macromolecules across the glomerular capillary wall.

TLRs

Cell surface and intracellular molecules of eukaryotic cells that recognize structurally conserved molecules derived from microorganisms, activating innate immunity serving as a first line of defence against foreign pathogens.

(NCT02682407)²²⁵, narsoplimab was well tolerated and resulted in a substantial reduction in proteinuria; however, this effect was observed in only a subset of patients, possibly because of the lower prevalence (20–30%) of lectin pathway activation in the white population²²⁶. Despite promising outcomes in the phase II trial, the interim analysis of the phase III trial yielded unfavourable results, leading to the discontinuation of the trial (NCT03608033)²²⁷.

As gut dysbiosis may be implicated in the pathogenesis of IgAN, modulation of gut microbiota may be a promising therapeutic approach. In a humanized mouse model of IgAN, treatment with antibiotics led to a reduction in IgA1-related immune complexes and mesangial IgA1 deposition^{228,229}. Another potential strategy is the use of probiotics or faecal microbiota transplantation to regulate gut microbiota^{230,231}. Encouraging results from case series have led to the initiation of a clinical trial evaluating the efficacy of the use of faecal microbiota transplantation in patients with IgAN (NCT05182775)^{232,233}.

Management of special patient populations

In patients with RPGN, the benefits of intensive immunosuppressive therapy outweigh the potential risks, as untreated patients may rapidly progress to kidney failure. There have been no randomized trials, but observational data support the use of the treatment regimens used in antineutrophil cytoplasmic antibody-associated vasculitis²³⁴. Regarding secondary IgAN, therapy should be directed at the underlying primary disease. There are few data on the treatment of IgAV. Steroid therapy should not be used to prevent kidney involvement and supportive therapy should be the main pillar in the treatment of IgAV without other systemic manifestations⁵⁴. An immunosuppressive approach might become necessary due to potentially life-threatening systemic manifestations^{235,236}.

The treatment regimen in children differs and must be considered separately by paediatric nephrologists^{237,238} (Box 2). The optimal therapy approach in children with IgAN is uncertain due to the lack of guidelines²³⁹. Risk assessment through the Oxford classification of IgAN MEST-C score and the IgAN prediction tool is crucial to determine appropriate treatment intensity¹⁶⁹. Supportive care should be provided for all children with IgAN and proteinuria of ≥ 0.2 g/day per 1.73 m² for at least 3–6 months before considering additional therapy. Glucocorticoid therapy is suggested for high-risk patients with persistent proteinuria of >0.5 g/day per 1.73 m² or severe kidney lesions, and is typically administered for 6 months. Prolonged glucocorticoid use is discouraged owing to potential adverse effects. Oral cyclophosphamide and glucocorticoids may be considered for children with RPGN; however, limited evidence supports their efficacy²³⁹. Treatment of recurrent IgAN after transplantation has not been systematically evaluated. Data from the ANZDATA registry suggest that continuation of corticosteroids might reduce graft loss in recurrent IgAN²⁴⁰, but most clinicians focus on optimizing supportive care. A potential role for SGLT2 inhibitors and TRF-budesonide is currently under investigation in this situation.

Kidney transplantation is the treatment of choice for patients with end-stage kidney disease, but recurrent IgA deposition in the transplant is very common (Box 3).

Quality of life

IgAN can considerably affect the quality of life of children and adults. However, only a few studies have systematically investigated the mental, social and physical costs of living with IgAN^{241–243}. Those studies that have been performed list a range of symptoms, some of which are probably treatment-related, reported by adults and children with IgAN

Box 3

Transplantation

IgA nephropathy (IgAN), along with focal segmental glomerulosclerosis and membranous nephropathy, is one of the most common glomerulonephritides that recur after transplantation²⁹⁶. Historically, this has been considered by clinicians to be unproblematic, as data had shown that the number of transplant failures in those with and without IgAN recurrence did not seem to differ during the first 10 years after transplantation²⁹⁷. Patients with IgAN have been reported to have the lowest rates of graft loss and mortality compared with patients who have recurrence of other glomerulonephritides, with the proportion of graft failures due to recurrence of IgAN at only 1.6%²⁹⁸. In general, however, rates of recurrent IgA deposition are considerably higher but often remain clinically silent. This is confirmed by the evaluation of protocol biopsies, in which half of the patients showed IgAN recurrence in the graft after 5 years²⁹⁹. Longer-term follow-up studies suggest that mortality and risk of graft failure due to recurrent IgAN increases 12–15 years after transplantation^{300,301}. IgAN transplant recipients with recurrent disease had double the risk of losing their allograft. In the post-transplant glomerular disease project (TANGO), which included an international cohort, there was a 32% graft loss at 8 years after diagnosis of recurrence, which was significantly higher than in patients with no recurrent IgA deposition³⁰². Thus, it is important to consider including MEST-C scoring for recurrent IgAN in assessment of allograft biopsy samples³⁰³.

including fatigue, weight gain and swelling, sleep problems, acne, hair loss, hirsutism, loin pain and reduced mental wellness. Patients often report that a number of these symptoms are episodic, leading to good and bad days and that the lack of control over the trajectory of their disease and symptoms leads to feelings of helplessness and frustration.

IgAN also negatively affects many day-to-day activities, including participation in hobbies, social and business travel and engagement in sports and group activities. As a consequence, patients commonly report feeling isolated, and this is particularly common in children and young adults who develop adverse effects of corticosteroid use that affect their appearance²⁴³. In addition, IgAN can negatively affect relationships with friends and family, as well as productivity at work, owing to mental and physical fatigue. It is, therefore, not uncommon for patients to report financial struggles due to the inability to work^{242,243}.

Outlook

As our pathophysiological understanding of IgAN has increased, more rational therapeutic options have emerged. While the evidence for genetic effects on disease manifestation and progression has become stronger, so has evidence for the relevance of a gut–kidney axis in driving IgAN. This has led to the approval of the first drug for use in IgAN. TRF-budesonide has the potential to finally offer an alternative to systemic corticosteroid therapy, which has long been considered the only immunomodulatory treatment for patients at high risk of progression. Although the efficacy of systemic corticosteroids has been widely reported, there is a consensus that the risk of serious adverse events

is unacceptably high and that safer and better tolerated therapies are needed.

The foundational treatment for all patients with IgAN is supportive care. Supportive care options are also increasing with the addition of SGLT2 inhibitors and endothelin receptor antagonists, such as sparsentan, and possibly mineralocorticoid receptor antagonists. Other new approaches such as complement inhibition and interference with B cell maturation and proliferation are also being evaluated in IgAN and will offer yet more treatment options.

It is gratifying to see that after years of stagnation, multiple new and emerging approaches are now available to treat patients with IgAN at risk of progression. With these new options, new questions emerge. First, do we need to alter the current approach of first optimizing supportive care before considering immunosuppression? With the advent of safer therapies, should we adopt an approach similar to that used in rheumatology, in which treatment should be intensive early-on to induce remission, followed by step-down therapy to target every salvageable nephron? These considerations lead to the second question: what is the best combination of therapies, and can biomarkers be identified to guide treatment choices and enable personalization of induction and maintenance of remission regimens? The third question might be the most challenging: will all these new therapies be affordable, both as single treatments and as combination treatments? Hence, identifying the best treatment combinations and biomarkers to guide therapeutic choices to enable improved personalized and cost-effective therapeutic regimens is an important task for the future.

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Author contributions

Introduction (R.K. and E.S.); Epidemiology (R.K., C.S. and E.S.); Mechanisms/pathophysiology (E.S., J.B. and J.F.); Diagnosis/screening/prevention (E.S., C.S., S.C.W.T., V.T., J.B., J.F. and P.B.); Management (E.S., C.S., S.C.W.T., V.T. and J.F.); Quality of life (J.B.); Outlook (J.F.); Overview of the Primer (R.K.).

Competing interests

S.C.W.T. has acted as a consultant for Boehringer Ingelheim, Eledon Pharmaceuticals and Trave, and has received speaker's honoraria from AstraZeneca, Bayer, Boehringer Ingelheim, GSK and Novartis. He is an Executive Committee member of KDIGO. P.B. has received consulting fees from Bristol Myers Squibb, lecture fees from Novo Nordisk, Novartis, Astellas and research funding from Ergonex Pharma GmbH and Affiris AG, but none in association with this work. V.T. has acted as a consultant for or has received honoraria from AstraZeneca, Bayer, Boehringer-Ingelheim, Calliditas, GSK, Novartis, Omeros, Otsuka and Trave. J.F. has acted as a consultant for or has received honoraria from AstraZeneca, Boehringer, Calliditas, Chinook, Novartis, Omeros, Roche, Stadapharm, Trave and Vifor. He was also chair of the data safety monitoring committee of a trial by Visterra. J.B. has acted as a consultant for or has received honoraria from Alnylam, Argenx, Astellas, BioCryst, Calliditas, Chinook, Dimerix, Galapagos, Novartis, Omeros, Trave Therapeutics, Vera Therapeutics and Visterra; has received grant support from Argenx, Calliditas, Chinook, Galapagos, GlaxoSmithKline, Novartis, Omeros, Trave Therapeutics and Visterra; and has been involved in the following clinical trials: ADU-CL-19 & ALIGN (Chinook), APPLAUSE (Novartis), ARTEMIS-IGAN (Omeros), ENVISION (Visterra), NefligARD (Calliditas) and ORIGIN (Vera Therapeutics). His research projects include Argenx, Calliditas, Chinook, Galapagos, GlaxoSmithKline, Novartis, Omeros, Trave Therapeutics, Visterra. R.K. is a founder, shareholder and board member of Sequantrix GmbH, a member of the scientific advisory board of Hybridize Therapeutics, has received honoraria for advisory boards and talks from Bayer, Chugai, Pfizer, Roche, Genentech, Lilly and GSK and has research funding from Trave Therapeutics, Galapagos, Novo Nordisk and Ask Bio. R.K. was supported by grants from the German Research Foundation (DFG; SFBTRR219: CRU344 428857858 and CRU5011 445703531), by two grants from the European Research Council (ERC-StG 677448, ERC-CoG 101043403), a grant from the Else Kroener Fresenius Foundation (EKFS), the Dutch Kidney Foundation (DKF), TASKFORCE EP1805 and Kolff Grant no. 113351, the NWO VIDI 09150172010072 and a grant from the Leducq Foundation, and the BMBF eMed Consortium Fibromap and the BMBF Consortium CureFib. C.S. has received honoraria from Stadapharm and is supported by the clinical research unit InteraKD consortium CRU5011 (project ID 45703531) of the DFG and by the Dr. Werner Jackstädt Stiftung. E.S. is supported by the clinical research unit InteraKD consortium CRU5011 (project ID 45703531) of the DFG and by STOP-FSGS by the German Ministry for Science and Education (STOP-FSGS-01GM2202C).

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